Near Field Electromagnetic Focusing using Dipoles

Fangwei Chang, Department of Electrical and Computer Engineering, University of Toronto

***Abstract*—This paper demonstrates near-field focusing of electric and magnetic fields using a linear array of dipoles. The system was implemented using a 2D FDTD scheme, with Gaussian target functions specified at a focal plane. Properties of focusing in dielectric materials was also investigated. This allowed the creation of sharply peaked fields, which can be used in a variety of areas such as wireless power transfer and medical applications.**

***Index Terms*—Beamforming, field focusing, FDTD, near field, diffraction limit, microwaves**

I. Introduction

FOCUSING of electromagnetic fields in the near field beyond the diffraction limit is of great use in microwave and optical research. Relevant technological applications can also be specified: examples include wireless power transfer (WPT), where sharply focused fields allow for greater transmission range between the Rx and Tx coils, as well as the medical field, where such fields can be used to heat or stimulate tissue in an non-invasive manner.

II. Analytic Solution in Vacuum

The solution to the focusing problem in vacuum can be solved analytically. The fields generated by the dipoles can be regarded as a non-orthogonal basis with which a target function can be reproduced at the focal plane. This is achieved by assigning multiplicative weights to the field of each dipole as well as an appropriate phase. Target functions in this scenario are Gaussian, achieved at a pre-determined distance away from a linear dipole array (the focal plane).

For magnetic fields generated by a wire loop, the following relations for H hold in the near field ($kr \ll 1$) [1]. Only the radial component H_r is considered as the primary goal is to achieve a forward-directed focal point.

$$H_r \sim \frac{a^2 I_o e^{-jkr}}{2r^3} cos\theta \tag{1}$$

$$H_\theta \sim \frac{a^2 I_o e^{-jkr}}{4r^3} sin\theta \tag{2}$$

$$H_\phi = E = 0 \tag{3}$$

For electric fields generated by a slot dipole antenna on the yz plane radiating in the x direction, the following result holds [2]:

$$E_x(x,y,z) = \frac{I_m y}{j2\pi} \frac{e^{-jk\sqrt{x^2+y^2+L^2/4}}}{\rho^2} \tag{4}$$

The one-to-one correspondence of the equation forms suggest that the focusing problem is essentially the same for H and E. For a focal plane/domain of N_x cells and N dipoles, the optimal weights w_{opt} is performed using a method of moments expansion and is given by the following [2]:

$$w_{opt} = (A^H A)^{-1} A^H \cdot b \tag{5}$$

TABLE I
Analytic weights for N = 9 Dipoles

Dipole 1	7.5245e+00 - 2.4664e-04i
2	-2.8384e+01 + 5.5581e-04i
3	6.7522e+01 - 1.5354e-03i
4	-2.5445e+02 + 5.9521e-03i
5	6.1065e+02 - 1.9542e-02i
6	-2.5444e+02 + 5.9520e-03i
7	6.7495e+01 - 1.5351e-03i
8	-2.8341e+01 + 5.5537e-04i
9	7.4822e+00 - 2.4603e-04i

Where b is the target function $f(x)$ at the focal plane, and A is an $N_x \times N$ array corresponding to the field generated at the focal plane by each of the N dipoles. The resulting vector of weights w_{opt} is complex.

Ultimately, the H_θ and H_ϕ components of the field do not contribute substantially enough to affect the solution; experimentation with the results showed that H_r alone sufficed. The solution was implemented on a 500 $\times$ 60 cell domain of dx = dz = 2.5mm. The target function $f(x)$ was specified as a Gaussian distribution centered at the midpoint of the dipole array, with $\sigma = x_{max}/20$.

Increasing the number of dipoles generally improved the focusing resolution. For a linear array of 9 dipoles, a significant reduction in beam narrowness was observed, with the 3 dB distance reduced from 12.5 cm (50 cells) to 8.75 cm (35 cells) (Fig. 3). Calculated weights for the array are specified; weights can be achieved by tuning currents in loops for magnetic fields (Table I).

III. FDTD Implementation with Dielectric

Focusing of the was implemented investigated using the 2D FDTD scheme. In two dimensions, electromagnetic effects are separated into the decoupled transverse magnetic (TM) and transverse electric (TE) modes. The problem is first implemented in the TM mode (E_z, H_x, H_y) using electric dipoles as sources, although the TE problem (H_z, E_x, E_y) is analogous. In the near field, magnetic and electric fields may be considered independently of each other.

A domain of $N_x \times N_y$ = 200 $\times$ 60 cells with dx = dy = 2.5 mm was specified (0.5 m x 0.15 m). The domain was terminated using perfectly matched layers (PML) of 10 cell thickness, resulting in a 180 x 40 cell interior region. The focal plane was set to the N_y = 40 axis. The source dipoles were uniformly distributed along the N_y = 10 axis in the

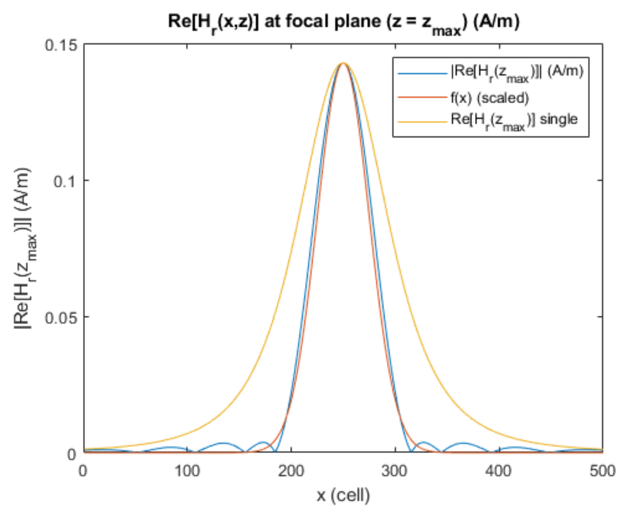

Fig. 1. Analytic focusing result for magnetic fields. Target function (orange) and achieved H field (blue) shown, as compared to narrowest achievable profile from single dipole (yellow). Distribution peaks are scaled to same magnitude for ease of comparison.

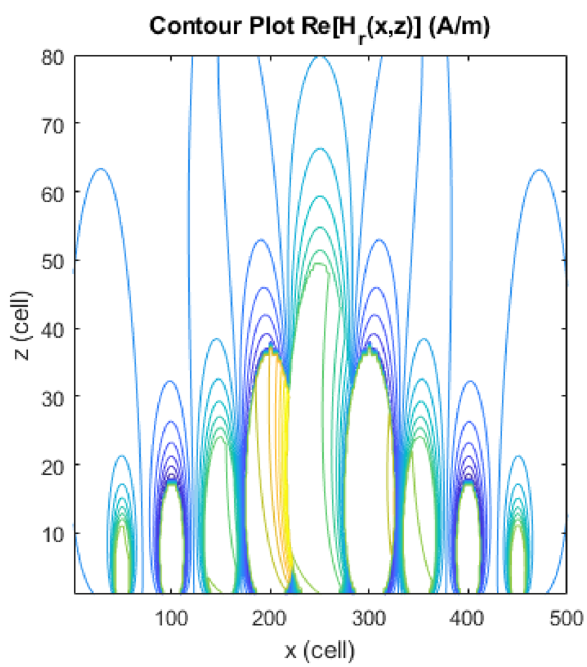

Fig. 2. Analytic focusing result for magnetic fields. Contour plot of focused field for 9 optimally weighted loops, with field amplitude cutoff at $|H_r|$ = 1.5 A/m. Phase, ie. imaginary component neglected.

interior, implemented as harmonic sources of the following form (dipole antennas for H, or slot antennas for E):

$$\vec{F}(\vec{r}) = cos(\omega t + \phi) \tag{6}$$

The simulation was performed at f = 10 kHz, a particularly low-frequency signal which places the problem in the static, near field regime. The phase ϕ corresponds to $arctan\frac{Re(w)}{Im(w)}$ of the dipole's assigned weight w. Using the analytic method outlined, w_{opt} is obtained.

A time step of dt = 4.1696 $\times 10^{-12}$ is used, corresponding to the stability requirement of dt = $\Delta x/2c$.

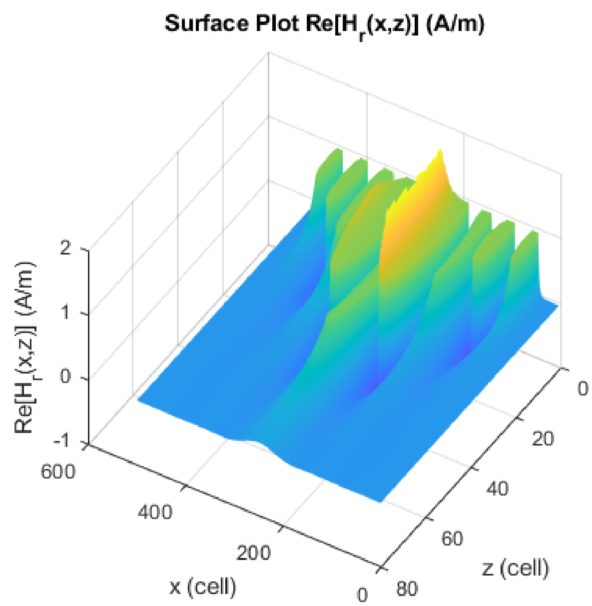

Fig. 3. Analytic focusing result for magnetic fields. Surface plot of focused field for 9 optimally weighted loops, with field amplitude cutoff at $|H_r|$ = 1.5 A/m. Phase, ie. imaginary component neglected.

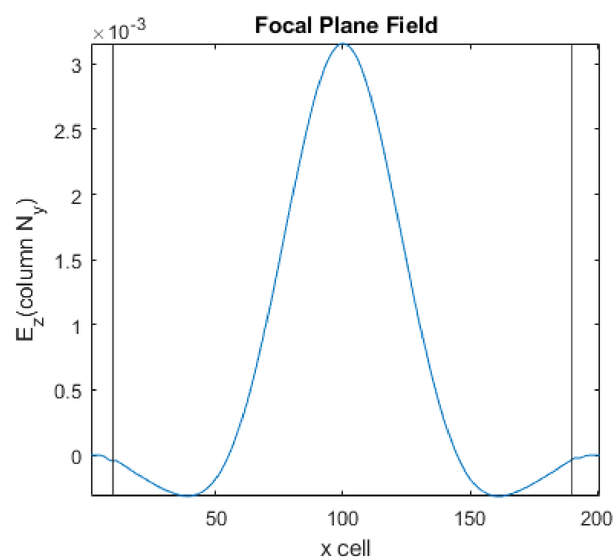

Fig. 4. Resulting focal field at corresponding focal plane for N = 3 dipole sources. Full width at half maximum (FWHM) corresponds to approximately 6.0 cm, or 24 cells of 2.5 mm each.

IV. Three-Dipole Array Focusing

Using the analytic solution for a vacuum medium, the corresponding weights for three dipoles were obtained for narrower Gaussian target function $f(x)$ centered at $x_{max}/2$ with $\sigma = x_{max}/40$. The resulting function is limited by the low number of dipoles used; the 3 dB half-distance of the peak corresponded to approximately 6.0 cm (24 cells), compared to 8.25 cm for a single dipole (33 cells). The result suggested that additional dipoles should be included (Fig. 4).

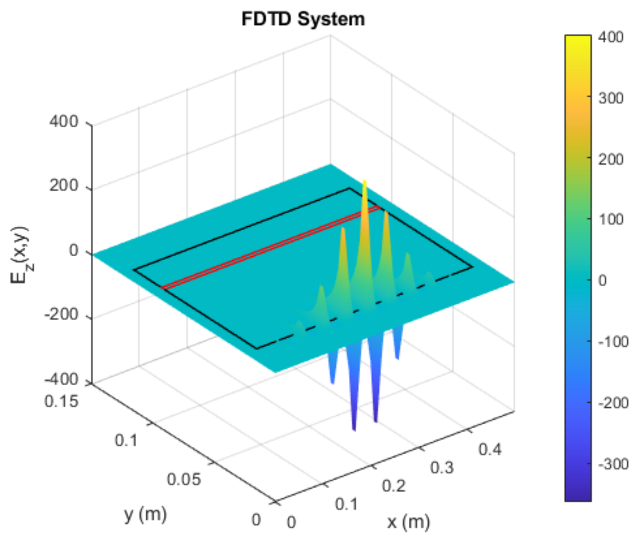

Fig. 5. Surface plot of FDTD system with weighted array of 19 dipoles, 10 kHz. Boundary between interior and PML marked by black border. Focal plane marked by red border. Symmetric arrangement of dipoles to replicate Gaussian target is evident.

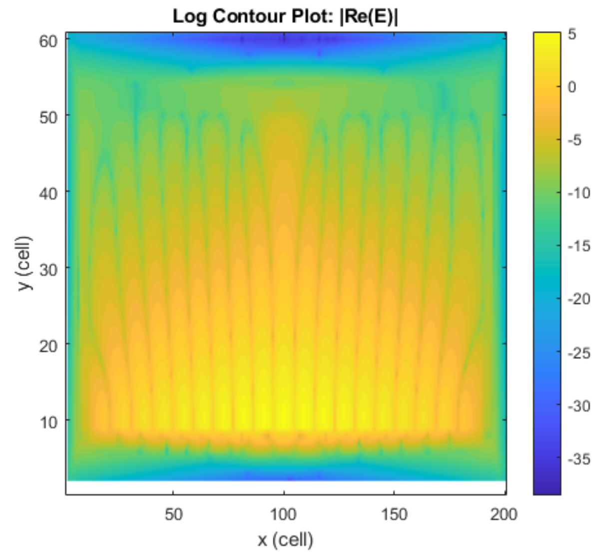

Fig. 6. Logarithmic contour of absolute field amplitude of weighted array of 19 dipoles, 10 kHz. Focal plane implemented at cell y = 40.

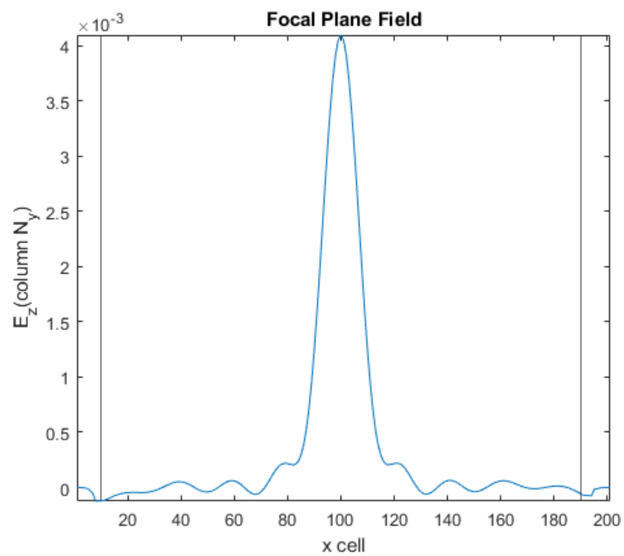

Fig. 7. Resulting focal field at corresponding focal plane for N = 19 dipole sources. Full width at half maximum (FWHM) corresponds to approximately 2.25 cm, or 9 cells of 2.5 mm each.

V. Nineteen-Dipole Array Focusing

To improve the resolution of focusing, the system was re-computed using $N = 19$ dipoles. An optimal method for determining the number of dipoles was not determined, although by inspection the addition of more sources becomes ineffective when their fields begin to strongly overlap (as per contour plot). The same Gaussian target function at y = 10 cm is retained from the $N = 3$ dipole system.

The phase of the dipoles were in this situation small but not inconsequential. With only the real component included, the sign of the peak field was observe to invert per approximately 1.6 $\times 10^4$ time steps (ie. $\sim$0.06 μs) as a result of the non-optimal phase. By accounting for the imaginary component, the focused field remains steady up to 10^5 simulated time steps.

The 3 dB half-distance of the peak for the 19 dipole arrangement was observed to be 2.25 cm (9 cells), compared to 8.25 cm for a single dipole (33 cells); this represents a substantial improvement in focusing quality (Fig. 7). The amplitude of the field at the focal plan is however reduced compared to the source field by up to a factor of 10^3, although the method for obtaining weights optimizes primarily for spatial accuracy as opposed to wave amplitude (Fig. 5, 6).

The size of the entire focusing setup is 0.5 m in length and achieves focusing at a distance of 0.1 m, dimensions which permits a reasonable physical realization.

VI. Effect of Dielectric Medium

At the original frequency of 10 KHz, the fields are virtually static and results in a scenario where the addition of pure dielectrics do not significantly affect the focusing. Fields can be reasonably considered quasi-static for frequencies of 1 KHz - 1 MHz, which was the domain investigated.

For a range of logarithmically spaced frequencies in the quasi-static range, the effect of dipole (source) frequency on the resulting focal pattern was also observed to be negligible. This was true of other dielectric patterns as well, such as increasing the overall ϵ_r of the FDTD interior region, or adding a dielectric "wall" between the source array and the focal plane.

A. Focusing in Dielectric

The relative permittivity ϵ_r of the focal region from y = 25 to y = 50 is increased. In general the sharpness of the focused

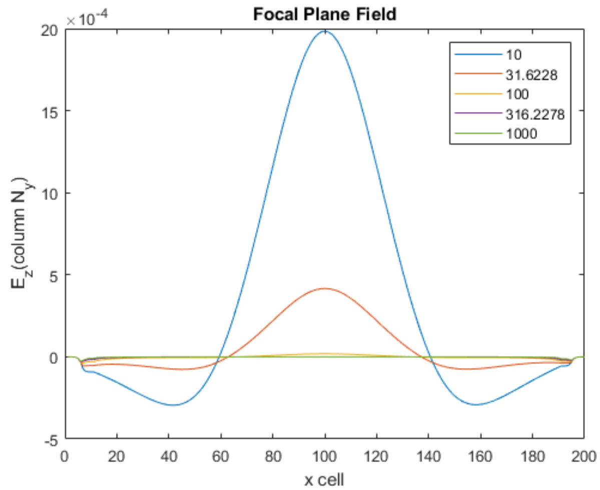

Fig. 8. Resulting focal field at corresponding focal plane for N = 3 dipole sources in materials of various permittivities ϵ_r (legend), stimulated for approximately 10^{12} time steps.

TABLE II
OPTIMAL WEIGHTS FOR N = 19 DIPOLES

Dipole 1	-1.1809e+00 + 1.0056e-03i
2	6.2746e+00 - 5.2857e-03i
3	-1.8425e+01 + 1.5451e-02i
4	4.0312e+01 - 3.3847e-02i
5	-7.4477e+01 + 6.2838e-02i
6	1.2411e+02 - 1.0565e-01i
7	-1.9275e+02 + 1.6633e-01i
8	2.7911e+02 - 2.4550e-01i
9	-3.6362e+02 + 3.2656e-01i
10	4.0149e+02 - 3.6444e-01i
11	-3.6346e+02 + 3.2643e-01i
12	2.7881e+02 - 2.4523e-01i
13	-1.9233e+02 + 1.6596e-01i
14	1.2361e+02 - 1.0521e-01i
15	-7.3967e+01 + 6.2383e-02i
16	3.9863e+01 - 3.3441e-02i
17	-1.8106e+01 + 1.5156e-02i
18	6.1119e+00 - 5.1309e-03i
19	-1.1373e+00 + 9.6177e-04i

pattern was unaffected, although the peak amplitude resulting from the optimization process was reduced (Fig. 8).

VII. CONCLUSION

The paper has demonstrated a method for focusing of electric and magnetic fields using a linear array of dipoles, implemented via 2D FDTD. The result allows for the formation of sharply focused beams in the near field region, which has wide-ranging utility in research and technological applications.

Future work, not implemented here due to time constraints, will extend the model to 3D as well as simulating the fields in more complex dielectric systems. Optimizing for an ideal set of parameters such as dipole number, dipole spacing and domain size also represents a priority. Lastly, more complex materials with a wide range of permittivities, conductivities and frequency response patterns should be investigated to determine the robustness of the system in practical applications.

Presently, the optimization for determining weights only considers the fields in a vacuum environment; it is possible to modify this scheme so as to render it more responsive to the operating environment/immediate surroundings.

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